CE 488/5588: Applied AI
Topic 14: Generative Deep Learning in Practice — GANs, VAEs, Diffusion, and Engineering Applications

Lecture Roadmap

From Prediction to Synthesis

Topic 13 focused on why deep learning became trainable and how major architectures emerged.

Topic 14 asks a different question:

- what do generative models actually do in practice,
- where are they being used in engineering,
- and what should we trust or distrust about their outputs?

This chapter is intentionally application-heavy.

Why Engineers Should Care

Generative AI is relevant to engineering because it can support:

- design-space exploration,
- scenario generation,
- data augmentation,
- reconstruction of missing or corrupted information.

Examples

Crack-mask synthesis, anomaly-sensitive monitoring, and floorplan generation are all generative tasks.

What Question Do Generative Models Answer?

Discriminative models ask:

- What class is this?
- What value should I predict?

Generative models ask:

- What realistic sample could have produced this observation?
- What plausible new sample could I synthesize?

Plausible Is Not the Same as Valid

Core Caveat

A generative model may produce outputs that look convincing while still violating physical laws, code requirements, or engineering constraints.

In engineering, realism is not enough. Validation remains essential.

What This Chapter Covers

- a short primer on generative modeling,
- GANs for crack and damage imagery,
- VAEs for SHM and monitoring,
- diffusion for media generation and design,
- comparison, limitations, and engineering judgment.

Discriminative versus Generative Thinking

Main Distinction

Discriminative models learn decision boundaries or input-output mappings. Generative models learn enough of the data distribution to synthesize, reconstruct, or interpolate samples.

For engineers, the distinction is practical rather than philosophical.

Engineering Use Cases for Generative Models

Generative models are useful when the task involves:

- synthesis of rare events,
- reconstruction or imputation,
- exploration of candidate designs,
- learning what “normal” looks like.

Three Main Families at a Glance

Three Influential Families

VAEs emphasize latent structure, GANs emphasize adversarial realism, and diffusion models emphasize iterative denoising with strong conditioning.

These families solve similar broad problems with very different trade-offs.

VAE in One Slide

- Encode data into a latent representation.
- Decode from that latent space back to data.
- Strength: smooth latent interpolation and stable training.
- Weakness: outputs may be overly smooth or blurry.

GAN in One Slide

- A generator proposes synthetic samples.
- A discriminator tries to detect whether they are fake.
- Strength: visually sharp outputs.
- Weakness: unstable training and mode collapse.

Diffusion in One Slide

Diffusion Idea

Add noise to data, then train a model to reverse that process step by step. Generation starts from noise and progressively becomes structured.

This is why diffusion is often explained as controlled denoising.

How Should Engineers Choose?

The right family depends on the objective:

- structured latent modeling and anomaly reasoning → VAE,
- sharp targeted synthesis under supervision → GAN,
- flexible conditioned generation → diffusion.

GANs: The Adversarial Intuition

At a high level, GAN training is a game:

- the generator tries to produce convincing samples,
- the discriminator tries to reject them,
- both improve in response to each other.

Why This Matters

Adversarial loss can preserve structure better than simple pixel-wise averaging, especially for thin, connected defects.

Why Crack Detection Is a Good GAN Testbed

Crack imagery is difficult because:

- cracks are thin and sparse,
- labels are often noisy,
- shadows, stains, and road texture create clutter,
- ordinary losses can blur or erase the target structure.

ICGA for Road-Surface Crack Detection

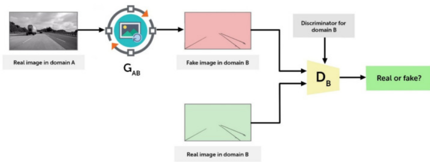


Figure: ICGA architecture for road-surface crack detection.

This is a strong example of using conditional adversarial learning inside a practical inspection pipeline.

Source: Road Surface Crack Detection Method Based on Conditional GANs

Why the ICGA Figure Matters

The figure is pedagogically useful because it shows that:

- engineering GANs are usually conditional,
- they are often paired with segmentation-style backbones,
- they are meant to improve structure, not generate arbitrary art.

CrackGAN and Imperfect Labels

CrackGAN addressed a realistic annotation problem:

- crack masks are often incomplete,
- the positive class is highly imbalanced,
- a model may collapse to "all background" if not properly regularized.

Source: CrackGAN (arXiv)

Why GANs Help in This Setting

Engineering Interpretation

GANs in civil infrastructure vision often work as structure-preserving regularizers. They encourage outputs that look globally plausible rather than merely minimizing pixel error.

That makes them attractive for connected crack morphology.

Conditional GAN Pavement Segmentation

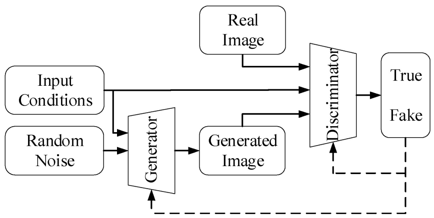


Figure: Conditional GAN structure for pavement-crack segmentation.

Here the generator maps a specific road image to a crack mask instead of generating free-form content.

Source: Pavement Cracks Segmentation Algorithm Based on CGAN

What This Tells Us About Engineering GANs

Common patterns in civil-engineering GAN adoption:

- pair GAN objectives with U-Net-style backbones,
- focus on segmentation and enhancement,
- use GANs to sharpen spatial coherence under field noise.

GAN Strengths in Civil Vision

Where GANs Help Most

- crack segmentation,
- image enhancement,
- data augmentation,
- structure-preserving reconstruction.

GAN Risks in Deployment

Why Better-Looking Is Not Enough

A GAN may hallucinate plausible continuity, sharpen false positives, or hide uncertainty. Visual quality alone does not guarantee decision quality.

This is especially serious in maintenance and safety applications.

Domain Shift Still Matters

Field deployment adds challenges:

- new cameras,
- different pavement textures,
- weather and lighting changes,
- seasonal wear and sealants.

Curated benchmark gains may not transfer cleanly.

GAN Takeaway for Infrastructure Vision

Main Takeaway

GANs are most valuable in civil vision when the target is thin, structured, and difficult to label precisely. Their role is less about art and more about preserving geometry under noisy supervision.

Why VAEs Fit Monitoring Problems

VAEs are often more useful in engineering monitoring than in photorealistic image synthesis.
They are good at:

- compact latent representation,
- interpolation across operating conditions,
- anomaly-style reasoning.

VAE Objective at a High Level

A VAE balances two goals:

- reconstruct the observed signal,
- organize the latent space so it stays smooth and sampleable.

Why Engineers Care

This makes VAEs useful for learning what normal variability looks like.

Mechanics-Informed Autoencoder Example

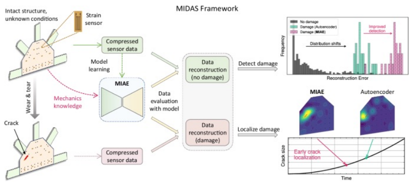


Figure: Mechanics-informed damage-detection and localization framework.

The pedagogical point is that the latent representation is tied to structural interpretation, not just compression.

Source: Mechanics-Informed Autoencoder Enables Automated Detection and Localization of Unforeseen Structural Damage

Why This Figure Is Important

It shows the direction of modern SHM research:

- latent models should reflect mechanics,
- autoencoders are more valuable when physically informed,
- damage localization is stronger than pure reconstruction alone.

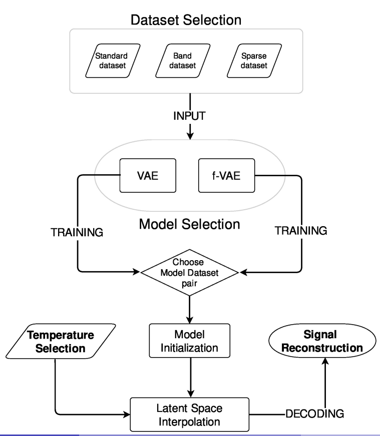
VAEs for Structural Health Monitoring

In SHM, the central challenge is often:

- damage affects the signal,
- but so do temperature, moisture, traffic, and operational conditions.

A useful latent model must represent ordinary variability without hiding damage.

Temperature Variation in Guided-Wave Monitoring



Why Temperature Is a Serious Engineering Problem

Temperature variation can:

- mimic damage,
- hide real damage,
- distort baseline comparisons,
- confuse anomaly thresholds.

This is why VAEs are compelling in monitoring settings.

What "Forced VAE" Means Conceptually

The main idea is not the exact formula.

Conceptual Lesson

Latent spaces do not become meaningful automatically. Engineering problems often require additional structure so that the latent manifold reflects real operational trends.

Strengths of VAEs in Monitoring

Best Uses

- baseline modeling,
- anomaly detection,
- interpolation across operating conditions,
- low-dimensional health-state representation.

VAE Failure Modes

Important Limitation

VAEs can oversmooth the signal. If trained carelessly, they may absorb weak early-stage faults as normal variability.

That makes domain knowledge and sensor physics essential.

VAE Takeaway for SHM

Main Takeaway

In engineering, VAEs are less about flashy generation and more about learning structured normality. Their real promise is disentangling nuisance variation from damage-sensitive behavior.

Why Diffusion Changed the Landscape

Diffusion models became dominant because they:

- train more stably than GANs,
- produce very high-quality outputs,
- support strong conditioning from prompts, masks, and layouts.

The Core Diffusion Intuition

Simple Story

Corrupt real data by gradually adding noise, then train a model to reverse that process. Generation starts from noise and repeatedly denoises until a coherent sample appears.

This is controlled iterative generation.

Why Diffusion Became Publicly Visible

The public saw diffusion through:

- text-to-image systems,
- viral prompt-based illustration,
- aesthetic filters and style transfer,
- later, text-to-video systems.

Social Implications of Diffusion Hype

Synthetic Media Changes Trust

Once realistic images and videos can be generated from prompts, visual evidence is no longer automatically trustworthy.

That matters for engineering communication and disaster response.

What New Risks Emerge?

- fabricated disaster imagery,
- deepfakes and identity misuse,
- text-to-video hype,
- rapid spread of persuasive but false visuals.

Why the Built Environment Fits Diffusion

Many built-environment problems are constrained generation problems:

- floorplans,
- site concepts,
- facade and interior layouts,
- design alternatives under partial specification.

Latent Diffusion for Floorplan Generation

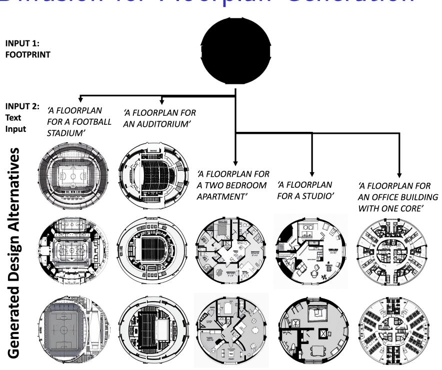


Figure: Latent-diffusion floorplan generation from footprint and text.

Why This Figure Works Pedagogically

It shows that diffusion is not only about artistic image synthesis.

- Inputs are structured and meaningful.
- Outputs are candidate design artifacts.
- The model supports rapid exploration rather than final approval.

DiffPlanner and Vector-Space Generation

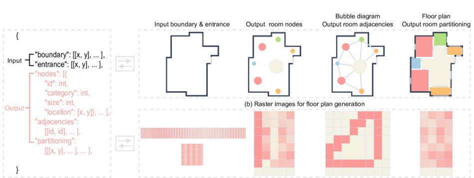


Figure: DiffPlanner: raster pipeline versus direct vector workflow.

This is especially relevant because engineering outputs are often structured geometry, not just images.

Source: Eliminating Rasterization: Direct Vector Floor Plan Generation with DiffPlanner

Why Vector-Space Generation Matters

Engineering Relevance

Raster generation may look plausible, but vector-space generation is much closer to real CAD/BIM-style workflows because it preserves geometry and topology more directly.

Human-in-the-Loop Design

Diffusion is best understood here as a proposal engine:

- propose candidate layouts,
- let humans refine and reject,
- use engineering judgment for compliance and feasibility.

Other Built-Environment Opportunities

Potential Uses

- early-stage design exploration,
- scenario visualization,
- concept generation,
- synthetic training data for rare cases.

Diffusion Risks in Engineering Use

Important Warning

A visually plausible floorplan is not automatically valid. Structural systems, egress, accessibility, MEP coordination, and code requirements still impose hard constraints.

Diffusion Takeaway

Main Takeaway

Diffusion models are powerful because they combine high-quality generation with conditioning. In engineering, their value is strongest when they support controlled exploration rather than unverified automation.

Three Families, Three Priorities

GANs

Realism and sharpness.

VAEs

Latent structure and smooth variation.

Diffusion

Controllable, high-quality generation.

Model Choice by Engineering Task

- Fast targeted synthesis with strong supervision → GANs.
- Latent interpolation and anomaly-style reasoning → VAEs.
- High-quality conditioned generation under prompts or layouts → diffusion.

Shared Challenges Across All Three

- satisfying physics and code constraints,
- evaluating quality beyond appearance,
- dataset bias and representativeness,
- compute cost and sustainability,
- trust and provenance.

Why Good-Looking Is Not Enough

Evaluation Challenge

Generative models can produce outputs that look convincing but are not useful, valid, or safe. Engineering quality requires task-specific evaluation, not just visual inspection.

Bias and Compute Are Engineering Issues Too

- Models inherit the biases of their training data.
- Large generative systems are expensive to train and deploy.
- Sustainability and resource cost are part of engineering judgment.

Trust and Provenance

Synthetic media quality is improving quickly.

What Changes?

Engineers, planners, and the public need stronger provenance, metadata, and verification practices because authentic-looking images are no longer guaranteed to be authentic.

Case Study Synthesis: Crack Inspection

GAN Lesson

In crack inspection, GANs are strongest when used to preserve thin defect structure under noisy labels and cluttered field imagery.

Case Study Synthesis: SHM and Monitoring

VAE Lesson

In monitoring, VAEs are strongest when used to model normal variability and separate nuisance factors such as temperature from potential damage-sensitive behavior.

Case Study Synthesis: Design Generation

Diffusion Lesson

In design, diffusion models are strongest when used to generate candidate layouts under conditioning, with humans still responsible for filtering, refining, and validating the results.

What Practitioners Should Remember

- Generative AI expands the design and search space.
- It does not replace engineering judgment.
- Validation, constraints, and domain knowledge remain the real workload.

Chapter Takeaways

- GANs, VAEs, and diffusion all address generative modeling, but with distinct trade-offs.
- Each family has a different engineering niche.
- The most valuable uses are controlled, task-specific, and human-supervised.

From Topic 14 to Topic 15

Diffusion made prompting and natural-language control visible to the public.
That leads to the next chapter:

- why text became a control interface,
- how language models work,
- and how the LLM era emerged from Transformer-based sequence modeling.

Final Message

Bottom Line

Generative AI is powerful not because it makes pretty outputs, but because it creates new ways to represent uncertainty, propose alternatives, and interact with engineering problems. The challenge is learning where that power is genuinely useful.